

# KARTHICK RAJA

DATA ANALYST · POWER BI DEVELOPER · BUSINESS INTELLIGENCE

Chennai, TN | +91-6383268445 | karthiikarthii46@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

## PROFESSIONAL SUMMARY

Data Analyst and Power BI Developer with hands-on experience in Business Intelligence, dashboard development, SQL, Python, ETL pipelines, and data visualization. Experienced in transforming large datasets (150K–650K+ records) into actionable insights through data modeling, DAX development, and interactive reporting. Skilled in Power BI, SQL, Python, and Excel with a strong focus on KPI reporting, business analytics, and data-driven decision-making.

## TECHNICAL SKILLS

- **Power BI & Business Intelligence:** Power BI, DAX, Power Query, Data Modeling, Power BI Service, KPI Development, Dashboard Development, Business Intelligence, Data Visualization, Data Storytelling, Stakeholder Reporting, Looker Studio
- **SQL & Databases:** SQL, MySQL, Joins, Aggregations, Subqueries, CTEs, Window Functions, Query Optimization
- **Python & Analytics:** Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Exploratory Data Analysis (EDA), Automation, Data Processing
- **Data Engineering & Modeling:** ETL Pipelines, Data Cleaning, Data Transformation, Star Schema Design, Data Validation, Data Governance
- **Excel & Reporting:** Excel, Pivot Tables, VLOOKUP, XLOOKUP, Conditional Formatting, Business Reporting, Reporting Automation
- **Cloud & Tools:** Microsoft Azure, Power BI Service, Azure Blob Storage, Git, GitHub, Streamlit

## PROFESSIONAL EXPERIENCE

### Data Analyst – Power BI Developer Intern

FEB 2026 – Present

Besant Technologies, Chennai

- Cleaned, validated, and transformed 600K+ records using Python (Pandas) and Excel — applying deduplication, null handling, outlier detection, and type standardization to reduce data quality errors by ~30% and deliver analytical-ready datasets for downstream reporting.
- Designed and deployed 5+ interactive Power BI dashboards featuring KPI cards, drill-through pages, dynamic slicers, bookmarks, and conditional formatting — enabling self-service analytics and KPI reporting for non-technical business stakeholders across Finance, Banking, and Sales domains.
- Developed 40+ optimized SQL queries including Joins, Aggregations, Subqueries, CTEs, and Window Functions supporting KPI reporting, trend analysis, and ad-hoc analytical requirements with output validated against source data.
- Built advanced DAX measures for MTD, QTD, YTD, Running Totals, SAMEPERIODLASTYEAR, MoM Variance %, and Growth Rate calculations — powering time-intelligence reporting and enabling period-over-period executive analysis across multiple business domains.
- Designed and executed end-to-end ETL pipelines using Python — data extraction, multi-step Pandas transformations, loading into MySQL, and post-load validation checks — ensuring reliable, accurate data models for Power BI consumption.
- Automated repetitive reporting and data-processing workflows using Python scripts — reducing manual effort by ~40% and improving operational efficiency across the analytics team.

## PROJECTS

### [Banking Performance Dashboard](#) · Power BI · SQL · DAX · Excel

- Analyzed 150K+ banking transactions (₹4.87B total transaction value) to monitor transaction volume, success rates, failure rates, tax collection, fees, and customer activity — designing a star-schema data model and building a Power Query ETL pipeline to transform raw MySQL data into analytical-ready datasets.
- Developed 20+ DAX measures — Transaction Success Rate %, Failure Rate %, Average Transaction Value, MTD/QTD/YTD Volume, MoM Growth %, and SAMEPERIODLASTYEAR comparisons using CALCULATE, SUMX, DIVIDE, and FILTER — enabling time-intelligence reporting across customer segments, regions, and transaction types.
- Wrote 15+ SQL queries using Joins, CTEs, Aggregations, and Window Functions (RANK, LAG) to extract segmented KPI datasets from MySQL and cross-validate Power BI aggregations against source records, ensuring 100% reporting accuracy.
- Delivered a 4-page interactive dashboard with conditional formatting thresholds (green/amber/red), state-wise transaction heatmap, drill-through failure analysis page, and dynamic bookmarks — replacing static monthly Excel reports and reducing report generation time from 3 days to on-demand access.
- Delivered executive-ready dashboards enabling stakeholders to monitor ₹4.87B transaction volume, identify failure-prone segments, and reduce manual reporting effort by approximately 80%.
- Conducted root-cause analysis on failed transaction patterns segmented by region, customer type, and transaction category, identifying 3 high-failure segments responsible for ~60% of operational errors and reducing manual investigation effort by ~35%.

- ▶ Designed a 5-page Power BI dashboard monitoring 50,000 IoT sensor readings across 10 industrial machines, tracking temperature, vibration, energy consumption, and predictive failure types
- ▶ Built 25+ DAX measures including time-intelligence (DATESMTD, DATESINPERIOD), anomaly rate %, machine health score, and rolling 7-day energy consumption for operational KPI tracking
- ▶ Engineered a Power Query ETL pipeline with 10 transformation steps: type casting, null handling, calculated date/hour columns, and machine status labeling for BMS-style IoT data
- ▶ Created an Anomaly Intelligence page with scatter-plot correlation (vibration vs temperature) to identify early failure signatures across machine categories
- ▶ Built an Energy & Efficiency page with hourly heatmap identifying peak consumption windows, directly replicating energy KPI requirements (kWh) for BMS/IoT analyst roles
- ▶ Implemented a Data Quality monitoring page tracking null sensor readings and out-of-range values per machine, enabling data pipeline health visibility.

## CERTIFICATIONS

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- ▶ PL-300 Microsoft Power BI Data Analyst
- ▶ Extract, Transform and Load Data in Power BI — Microsoft
- ▶ Harnessing the Power of Data with Power BI — Microsoft
- ▶ GenAI for Data Analytics — WsCube Tech
- ▶ Data Analyst Certification — Besant Technologies

## EDUCATION

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### **B.E. Computer Science & Engineering | 2022 – 2026**

Hindusthan Institute of Technology, Coimbatore | CGPA: 7.2/10